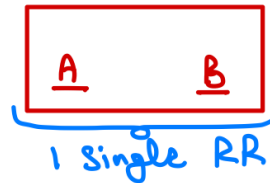


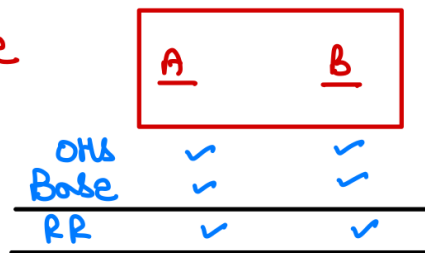
ACTIVITY BASED COSTING

1. Background of ABC

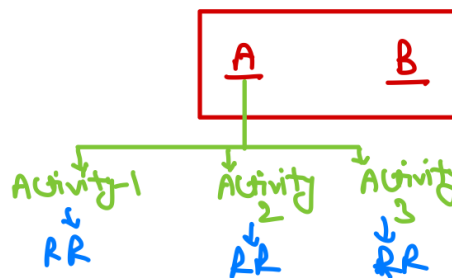
A) Blanket Rate



B) Departmental Rate



C) Activity Based



2. Cost Pool

It is the total cost of an activity.

3. Cost Driver

It is the base due to which cost changes

4. Steps in ABC

(A) Identify different activities

(B) Identify overheads related to activities

(C) Identify cost drivers

(D) Calculate activity cost driver rate (ACDR)

$$ACDR = \frac{\text{Budgeted Overheads of activity}}{\text{Budgeted Cost Driver}}$$

(E) Recover overheads based on ACDR

Trad. Method $\Rightarrow$ RR = $\frac{475020 + 579988 + 504992}{24000 + 54000} = \text{Rs. } 20 \text{ p.m.h.}$

Question – 1

PQR Pens Ltd. manufactures two products – ‘Gel Pen’ and ‘Ball Pen’. It furnishes the following data for the year 2017:

Product	Annual Output (Units)	Total Machine Hours	Total number of Purchase orders	Total number of set-ups
Gel Pen	5,500	24,000	240	30
Ball Pen	24,000	54,000	448	56

The annual overheads are as under:

Particulars		₹
Volume related activity costs	24:54	4,75,020
Set up related costs	30:56	5,79,988
Purchase related costs	240:448	5,04,992

Calculate the overhead cost per unit of each Product – Gel Pen and Ball Pen on the basis of:

- Traditional method of charging overheads $\frac{20 \times 24000}{5500} = 87.27 \rightarrow \frac{20 \times 54000}{24000} = 45$
- Activity based costing method and
- Find out the difference in cost per unit between both the methods. (a) Cost distortion (b) under/over costed

Solution**(i) Calculation of cost under Traditional Approach:**

Overheads rate per Machine hour = $\frac{\text{Total overheads}}{\text{Total machine hours}} = \frac{15,60,000}{24,000 + 54,000} = \text{₹}20 \text{ per machine hour}$

Statement of Cost

Particulars	Gel Pen	Ball Pen
Overheads absorbed (A)	$20 \times 24,000 = 4,80,000$	$20 \times 54,000 = 10,80,000$
Units (B)	5,500	24,000
Overheads per unit (A ÷ B)	87.27	45

(ii) Statement showing Activity Based Cost

Activity Cost Pool	Cost Driver	Ratio	Total Amount (₹)	Gel Pen (₹)	Ball Pen (₹)
Volume Related Activity Costs	Machine Hour	24:54	4,75,020	1,46,160	3,28,860
Set-up Related Costs	No. of Set-ups	30:56	5,79,988	2,02,321	3,77,667
Purchase Related Costs	No. of Purchase Orders	240:448	5,04,992	1,76,160	3,28,832
Total Costs				5,24,641	10,35,359
Output (Units)				5,500	24,000
Cost per unit				95.39	43.13

(iii) Statement of Difference in Cost

Particulars	Gel Pen	Ball Pen
Overheads cost per unit (₹) – Traditional Approach	87.27	45
Overheads Cost per unit (₹) – ABC	95.39	43.13
Difference per unit (Trad. - ABC)	-8.12	+1.87

under costed over costed

Question – 2

SK Ltd. manufactures three types of products namely P, Q and R. The data relating to a period are as under:

Total M.Nr. = $(10 \times 3000) + (12 \times 5000) + (14 \times 20000) = 41$ Total OH = $41 \times 6 = 246$

Particulars	P	Q	R
Machine hours per unit	$6 \times 10 = 60$	$6 \times 18 = 108$	$6 \times 14 = 84$
Direct Labour hours per unit	$20 \times 4 = 80$	$20 \times 12 = 240$	$20 \times 8 = 160$
Direct Material per unit (₹)	90	80	120
Production (units)	3,000	5,000	20,000

Currently the company uses traditional costing method and absorbs all production overheads on the basis of machine hours. The machine hour rate of overhead is ₹6 per hour. Direct labour hour rate is ₹20 per hour.

No. of Batches $\frac{3000}{150} = 20$ $\frac{5000}{500} = 10$ $\frac{20000}{1000} = 20$

The company proposes to use activity based costing system and the activity analysis is as under:

Particulars	P	Q	R
Batch size (units)	150	500	1,000
Number of purchase orders per batch	$20 \times 3 = 60$	$10 \times 10 = 100$	$20 \times 8 = 160$
Number of inspections per batch	$20 \times 5 = 100$	$10 \times 4 = 40$	$20 \times 3 = 60$

The total production overheads are analyzed as under:

Machine set up costs	$\rightarrow 20:10:20$	$\rightarrow 20\% \times 246 = 4.92$	50 batches 41 M.Nr. 200 Insp. 320 P.O. ACAP 9600 Per Batch 1.80 Per M.Nr. 4800 Per Insp. 750 Per P.O.
Machine operations costs	$\rightarrow 30:40:280$	$\rightarrow 30\% \times 246 = 7.38$	
Inspection costs	$\rightarrow 100:40:60$	$\rightarrow 40\% \times 246 = 9.84$	
Material procurement related costs	$\rightarrow 60:100:160$	$\rightarrow 10\% \times 246 = 2.46$	

Required:

P = 230 Q = 236 R = 364

- Calculate the cost per unit of each product using traditional method of absorbing all production overheads on the basis of machine hours.
- Calculate the cost per unit of each using activity based costing principles.

Solution

(i) Statement showing cost per unit – Traditional Method

Particulars	P	Q	R
Direct material	90	80	120
Direct labour [(4, 12, 8 hours) × ₹20]	80	240	160

Production overheads [(10, 18, 14 hours) × ₹6]	60	108	84
Cost per unit	230	428	364

(ii) Total machine hours = $(3,000 \times 10) + (5,000 \times 18) + (20,000 \times 14) = 4,00,000$

Total production overheads = $4,00,000 \times ₹6 = ₹24,00,000$

	Particulars	P	Q	R	Total
A	Production (units)	3,000	5,000	20,000	
B	Batch size (units)	150	500	1,000	
C	Number of batches (A ÷ B)	20	10	20	50
D	Number of purchase order per batch	3	10	8	
E	Total purchase order (C × D)	60	100	160	320
F	Number of inspections per batch	5	4	3	
G	Total inspections (C × F)	100	40	60	200

Statement of cost driver rate

Activity	Overhead (₹)	Cost driver quantity	Cost driver rate (₹)
Setup	$24,00,000 \times 20\% = 4,80,000$	$\left(\frac{3000}{150}\right) + \left(\frac{5000}{500}\right) + \left(\frac{20000}{1000}\right)$ batches	9,600 per batch
Inspection	$24,00,000 \times 40\% = 9,60,000$	200 inspections	4,800 per inspection
Purchases	$24,00,000 \times 10\% = 2,40,000$	320 purchases	750 per purchase
Machine operations	$24,00,000 \times 30\% = 7,20,000$	4,00,000 machine hours	1.80 per machine hour

Statement showing cost per unit – Activity Based Costing Method

Particulars	P	Q	R
Production units	3,000	5,000	20,000
	(₹)	(₹)	(₹)
Direct material (90, 80, 120)	2,70,000	4,00,000	24,00,000
Direct labour (80, 240, 160)	2,40,000	12,00,000	32,00,000
Machine related costs [(30,000, 90,000, 2,80,000) × ₹1.80]	54,000	1,62,000	5,04,000
Setup costs [(20, 10, 20) × ₹9,600]	1,92,000	96,000	1,92,000
Inspection cost [(100, 40, 60) × ₹4,800]	4,80,000	1,92,000	2,88,000
Purchase related costs [(60, 100, 160) × ₹750]	45,000	75,000	1,20,000
Total Costs	12,81,000	21,25,000	67,04,000
Cost per unit (Total cost ÷ Units)	427	425	335.20

$$\text{Trad. Rf} = \frac{2.700}{72000} = 3.75$$

$$800 \times 3.75 = 3000$$

$$600 \times 3.75 = 2250$$

Question – 3

GST Limited is a multi-product company. The production and cost details of its two products P and Q are given as follows:

Particulars	Product	
	P	Q
Quantity produced (No.)	9,000	7,200
Direct material cost (₹)	72,000	50,000
Direct labour hours	800	600
Purchase requisition (No.)	180	144
Production runs (No.)	144	108
Quality inspections (No.)	27	18

Direct wages rate is ₹14.50 per hour. Presently the company uses a single overhead recovery rate based on direct labour hours. Overhead incurred by the company during the year 2023-24 are as follows:

✓ Technical staff salary	₹45,000
✓ Machine operation expenses	₹1,62,000
✓ Machine maintenance expenses	₹27,000
Wages and salary of stores staff	₹36,000

During this period direct labour hours worked 72,000.

Now the Company wants to adopt Activity Based Costing. For this purpose, following activities are identified:

- ① • Quality control
- ② • Setup of machine for production runs
- ③ • Store receiving

It is also decided that salary of technical staff should be distributed among machine maintenance, setup and quality control in the ratio of 1 : 2 : 2. Machine maintenance expenses and machine operation expenses should be distributed in the ratio of 2 : 3 in between stores and production setup activities.

During this period cost drivers for these activities are identified as under:

- Requisition raised 5,760
- Production setup 7,200
- No. of quality test 720

You are required to compute:

- (i) The cost of products P and Q based on traditional absorption costing system.
- (ii) The cost of products P and Q based on ABC Costing system.

Solution

(i) Statement Showing “Total Cost - Traditional Method”

Particulars of Costs	P	Q
	(₹)	(₹)
Direct Materials	72,000	50,000
Direct Labour [(800,600 hours) × ₹14.5]	11,600	8,700
Production Overheads [(800,600 hours) × ₹3.75] (WN1)	3,000	2,250

Total Cost	86,600	60,950
Cost per unit (9,000, 7,200)	9.62	8.47

WN1: Calculation of Production Overhead:

	(₹)
Technical staff salary	45,000
Machine operation expenses	1,62,000
Machine maintenance expenses	27,000
Wages and salary of stores staff	36,000
Total Production Overhead	2,70,000
Total direct labour hours worked	72,000 hours
Production Overhead rate per hour	3.75

(ii) Statement Showing “Total Cost - Activity Based Costing”

Products	P	Q
Production (units)	9,000	7,200
	(₹)	(₹)
Direct Materials	72,000	50,000
Direct Labour [(800,600 hours) × ₹14.5]	11,600	8,700
Requisition Related Costs @ ₹20 per requisition raised (180,144) (WN2)	3,600	2,880
Production Setup Costs @ ₹19 per production runs (144,108) (WN2)	2,736	2,052
Quality Inspection Costs @ ₹25 per quality test (27,18) (WN2)	675	450
Total Cost	90,611	64,082
Cost per unit (9,000, 7,200)	10.07	8.90

WN2: Statement Showing Distribution of Expenses

	Machine maintenance expenses	Total Stores	Production Setup	Quality Control
Technical staff salary of ₹45,000 (1 : 2 : 2)	9,000	—	18,000	18,000
Machine operation expenses of ₹1,62,000 (2 : 3)	—	64,800	97,200	—
Machine maintenance expenses of ₹36,000 (2 : 3)	(9,000)	14,400	21,600	—
Wages and salary of stores staff	—	36,000	—	—
Total	—	1,15,200	1,36,800	18,000

WN3: Cost for each activity cost driver:

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Stores Receiving	1,15,200	5,760 Requisitions Raised	₹20 per requisition raised
Production Setup	1,36,800	7,200 Production Setup	₹19 per production setup
Quality Control	18,000	720 Quality Test	₹25 per quality test

$$\Delta L \text{ Inv.} = (4000 \times \frac{30}{60}) + (3000 \times \frac{45}{60}) + (1600 \times \frac{60}{60}) = 2000 + 2250 + 1600 = 5850$$

Question – 4

SK Ltd. is a multiproduct company, manufacturing three products S, K and M. the budgeted costs and production for the year ending March 31st, are as follows:

	<u>S</u>	<u>K</u>	<u>M</u>
Production quantity (units)	4,000	3,000	1,600
Resources per unit:			
Direct Material (Kg)	4	6	3
Direct labour (Minutes)	30	45	60

The budgeted direct labour rate was ₹10 per hour and the budgeted materials cost was ₹2 per Kg. Production overheads were budgeted at ₹99,450 and were absorbed to products using the direct labour hour rate. SK Ltd. followed an Absorption Costing System. $\rightarrow \frac{99450}{5850} = 17$

SK Ltd. is in consideration to adopt an Activity Based Costing System. The following additional information is made available for this purpose:

(a) Budgeted overheads were analyzed into the following:

Material handling	{	₹29,100 ÷ 38800 =
Storage costs		₹31,200 ÷ 30 =
Electricity		₹39,150 ÷ 36200 =

(b) The cost drivers identified were as follows:

Material handling	16000 : 18000 : 4800	→ Weight of material handled → 38800
Storage costs	10 : 5 : 15	→ Number of batches of material → 30
Electricity	24000 : 9000 : 3200	→ Number of machine operators → 36200

(c) Data on Cost Drivers was as follows:

	<u>S (4000)</u>	<u>K (3000)</u>	<u>M (1600)</u>
For complete production:			
Batches of material	→ 10	5	15
Per unit of production:			
Number of machine operators	→ 6	3	2

You are required to:

- Prepare a statement for management showing the unit costs and total costs of each product using the absorption costing method.
- Prepare a statement for management showing the products costs of each product using the ABC approach

(c) What are the reasons for the different product costs under the two approaches?

Solution

(a) Total Labour hours = $(4,000 \times 0.5) + (3,000 \times 0.75) + (1,600 \times 1) = 5,850$ hours

$$\text{Overheads per labour hour} = \frac{\text{Overheads}}{\text{Labour hours}} = \frac{99,450}{5,850} = \underline{\underline{\text{₹17 per labour hour}}}$$

Statement of Cost

Particulars	Product S	Product K	Product M
Direct Material	$4 \times 2 = 8$	$6 \times 2 = 12$	$3 \times 2 = 6$
Direct Labour	$0.5 \times 10 = 5$	$0.75 \times 10 = 7.50$	$1 \times 10 = 10$
Overheads	$0.5 \times 17 = 8.50$	$0.75 \times 17 = 12.75$	$1 \times 17 = 17$
Total Cost per unit	<u>21.50</u>	<u>32.25</u>	<u>33</u>
Units →	4,000	3,000	1,600
Total Cost →	86,000	96,750	52,800

(b) Material handling per kg = $\frac{\text{Material handling cost}}{\text{Weight of material handled}} = \frac{29,100}{(4,000 \times 4) + (3,000 \times 6) + (1,600 \times 3)} = \frac{29,100}{38,800} = \underline{\underline{\text{₹0.75}}}$
per kg

$$\text{Storage cost per batch} = \frac{\text{Storage cost}}{\text{No. of batches of material}} = \frac{31,200}{10+5+15} = \text{₹1,040 per batch}$$

$$\text{Electricity cost per machine operation} = \frac{\text{Electricity cost}}{\text{No. of machine operations}}$$

$$= \frac{39,150}{(4,000 \times 6) + (3,000 \times 3) + (1,600 \times 2)} = \frac{39,150}{36,200} = \underline{\underline{\text{₹1.08149 per machine operation}}}$$

Statement of Cost

Particulars	Product S	Product K	Product M
Direct Material →	$4 \times 2 = 8$	$6 \times 2 = 12$	$3 \times 2 = 6$
Direct Labour →	$0.5 \times 10 = 5$	$0.75 \times 10 = 7.50$	$1 \times 10 = 10$
<u>Overheads</u>			
Material handling →	$0.75 \times 4 = 3$	$0.75 \times 6 = 4.50$	$0.75 \times 3 = 2.25$
Storage cost →	$\frac{1,040 \times 10}{4,000} = 2.60$	$\frac{1,040 \times 5}{3,000} = 1.73$	$\frac{1,040 \times 15}{1,600} = 9.75$
Electricity cost →	$1.08149 \times 6 = 6.49$	$1.08149 \times 3 = 3.24$	$1.08149 \times 2 = 2.16$
Total cost per unit	25.09	28.97	30.16
Units	4,000	3,000	1,600
Total Cost	1,00,360	86,910	48,256

(c) Difference in total cost under two approach is due to difference in overheads borne by each department.

Question – 5

PQR Ltd. is engaged in the production of three products P, Q and R. The company calculates Activity Cost Rates on the basis of Cost Driver capacity which is provided as below:

Activity	Cost Driver	Cost Driver Capacity	Cost (₹)	ACDR
Direct Labour Hours	Labour Hours	30,000 Labour Hours	3,00,000	₹. 10
Production runs	No. of Production runs	600 Production runs	1,80,000	₹. 300
Quality Inspections	No. of Inspections	8000 Inspections	2,40,000	₹. 30

The consumption of activities during the period is as under:

Activity/ Products	P	Q	R	
Direct Labour hours	10x 10,000	10x 8,000	10x 6,000	→ 2.40l
Production runs	300x 200	300x 180	300x 160	→ 1.62l
Quality Inspection	30x 3,000	30x 2,500	30x 1,500	→ 2.10l

You are required to:

- Compute the cost allocated to each Product from each Activity.
- Calculate the cost of unused capacity for each activity
- A potential customer has approached the company for supply of 12,000 units of net product 'S' to be delivered in lots of 1,500 units per quarter. This will involve an initial design cost of ₹30,000 and per quarter production will involve the following:

Direct Material	₹18,000
Direct Labour hours	1,500 hours
No. of Production runs	15
No. of Quality Inspection	250

DM	18000
D. Exp. (30000 x 1500)	37500
DL (1500 x 10)	15000
P. R (15 x 300)	4500
Qual. (250 x 30)	7500
	48750
TC	
(+) Lot. Sale	58500

Prepare cost sheet segregating direct and indirect cost and compute the sales value per quarter of product 'S' using ABC system considering a markup of 20% on cost.

Solution**(i) Statement of Cost Driver Rate**

Activity	Amount (A)	Cost driver (B)	Cost Driver Rate (A÷B)
Direct Labour Hours	3,00,000	30,000 Labour Hours	✓ ₹10 per labour hour
Production runs	1,80,000	600 Production runs	✓ ₹300 per production run
Quality Inspections	2,40,000	8000 Inspections	✓ ₹30 per inspection

Statement of Cost

Particulars	P	Q	R	Total
Direct labour hour	10 × 10,000 = 1,00,000	10 × 8,000 = 80,000	10 × 6,000 = 60,000	2,40,000
Production run	300 × 200 = 60,000	300 × 180 = 54,000	300 × 160 = 48,000	1,62,000

Quality inspection	$30 \times 3,000$ = 90,000	$30 \times 2,500$ = 75,000	$30 \times 1,500$ = 45,000	<u>2,10,000</u>
Total Cost	2,50,000	2,09,000	1,53,000	6,12,000

(ii) Statement of Cost of Unused Capacity

Activity	Total Cost	Cost Charged to Products	Unused Cost
Direct Labour Hours	3,00,000	2,40,000	60,000
Production runs	1,80,000	1,62,000	18,000
Quality Inspections	2,40,000	2,10,000	30,000

(iii) Statement of Cost

Particulars	Amount (₹)
Direct material	18,000
Direct expenses (design cost) $\left(\frac{30,000}{12,000} \times 1,500\right)$	3,750
Prime Cost	21,750
Add: Overheads	
Direct labour hours $(1,500 \times 10)$	15,000
Production run (15×300)	4,500
Quality inspection (250×30)	7,500
COS	48,750
Add: Profit $(48,750 \times 20\%)$	9,750
Sales	58,500

Question – 6

SK is a global brand created by SK Ltd. The company manufactures three range of beauty soaps i.e. SK-Gold, SK-Pearl, and SK-Diamond. The budgeted costs and production for the month of March, 2021 are as follows:

	SK-Gold		SK-Pearl		SK-Diamond	
Production of soaps (Units)	4,000 $108 \times 4000 = 4.32l$		3,000 $106 \times 3000 = 3.18l$		2,000 $117 \times 2000 = 2.34l$	
Resources per Unit:	Qty	Rate	Qty	Rate	Qty	Rate
- Essential Oils	60 ml ⁴⁸	₹200 / 100 ml	55 ml ⁴⁴	₹300 / 100 ml	65 ml ⁵²	₹300 / 100 ml
- Cocoa Butter	20 g	₹200 / 100 g	20 g	₹200 / 100 g	20 g	₹200 / 100 g
- Filtered Water	30 ml	₹15 / 100 ml	30 ml	₹15 / 100 ml	30 ml	₹15 / 100 ml
- Chemicals	10 g	₹30 / 100 g	12 g	₹50 / 100 g	15 g	₹60 / 100 g
- Direct Labour	30 Min.	₹10 / hour	40 Min.	₹10 / hour	60 Min.	₹10 / hour

SK Ltd. followed an Absorption Costing System and absorbed its production overheads, to its products using direct labour hour rate, which were budgeted at ₹1,98,000.

$$DL \text{ hrs.} = \frac{(4000 \times 30)}{60} + \frac{(3000 \times 40)}{60} + \frac{(2000 \times 60)}{60} = 6000$$

$$\Rightarrow OH \text{ Rf} = \frac{1,98,000}{6000} = 33$$

Now, SK Ltd. is considering adopting an Activity Based Costing system. For this, additional information regarding budgeted overheads and their cost drivers is provided below:

Particulars	(₹)	Cost drivers
Forklifting cost →	58,000	Weight of material lifted $4.328 : 3.188 : 2.342 \Rightarrow 9.841$
Supervising cost →	60,000	Direct labour hours $2000 : 2000 : 2000 \Rightarrow 6000$
Utilities →	80,000	Number of Machine operations $20000 : 15000 : 12000 \Rightarrow 47000$

The number of machine operators per unit of production are 5, 5, and 6 for SK-Gold, SK-Pearl, and SK-Diamond respectively.

(Consider (i) Mass of 1 litre of Essential Oils and Filtered Water equivalent to 0.8 kg and 1 kg respectively (ii) Mass of output produced is equivalent to the mass of input materials taken together.)

You are requested to:

- Prepare a statement showing the unit costs and total costs of each product using the absorption costing method.
- Prepare a statement showing the product costs of each product using the ABC approach.
- State what are the reasons for the different product costs under the two approaches?

Solution

(i) Statement of calculation of labour hours

	SK- Gold	SK- Pearl	SK- Diamond	Total
Prod. of soaps (units) (A)	4,000	3,000	2,000	9,000
Direct labour (min.) (B)	30	40	60	-
Direct labour hours [(A×B)÷60]	2,000	2,000	2,000	6,000

Overhead rate per direct labour hour = $\frac{\text{Budgeted overheads}}{\text{Budgeted labour hours}} = \frac{1,98,000}{6,000} = ₹33$ per direct labour hour

Statement of cost

	SK - Gold	SK - Pearl	SK - Diamond
Essential oils	→ $\frac{200 \times 60}{100} = 120$	$\frac{300 \times 55}{100} = 165$	$\frac{300 \times 65}{100} = 195$
Cocoa Butter	→ $\frac{200 \times 20}{100} = 40$	$\frac{200 \times 20}{100} = 40$	$\frac{200 \times 20}{100} = 40$
Filtered water	→ $\frac{15 \times 30}{100} = 4.50$	$\frac{15 \times 30}{100} = 4.50$	$\frac{15 \times 30}{100} = 4.50$
Chemicals	→ $\frac{30 \times 10}{100} = 3$	$\frac{50 \times 12}{100} = 6$	$\frac{60 \times 15}{100} = 9$
Material cost per unit	→ 167.50	215.50	248.50
Direct labour per unit	→ $\frac{10 \times 30}{60} = 5$	$\frac{10 \times 40}{60} = 6.67$	$\frac{10 \times 60}{60} = 10$
Overheads per unit	→ $\frac{33 \times 30}{60} = 16.50$	$\frac{33 \times 40}{60} = 22$	$\frac{33 \times 60}{60} = 33$
Total cost per unit	→ 189.00	244.17	291.50
Number of units	4,000	3,000	2,000

Total costs	7,56,000	7,32,510	5,83,000
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(ii) Calculation of Cost Driver

Activity	Amount(₹) (A)	Cost driver quantity (B)	Cost Driver Rate (A ÷ B)
Forklifting	58,000	Gold – $[(60 \times 0.8) + 20 + 30 + 10] \times 4,000 = 4,32,000$ Pearl – $[(55 \times 0.8) + 20 + 30 + 12] \times 3,000 = 3,18,000$ Diamond – $[(65 \times 0.8) + 20 + 30 + 15] \times 2,000 = 2,34,000$ Total weight = 9,84,000	0.06 per gram
Supervising	60,000	Gold – $\frac{4,000 \times 30}{60} = 2,000$ Pearl – $\frac{3,000 \times 40}{60} = 2,000$ Diamond – $\frac{2,000 \times 60}{60} = 2,000$ Total machine hours = 6,000	10 per machine hour
Utilities	80,000	Gold – $5 \times 4,000 = 20,000$ Pearl – $5 \times 3,000 = 15,000$ Diamond – $6 \times 2,000 = 12,000$ Total operations = 47,000	1.70 per machine operation

Statement of cost

	SK – Gold	SK – Pearl	SK – Diamond
Material cost per unit	167.50	215.50	248.50
Direct labour per unit	$\frac{10 \times 30}{60} = 5$	$\frac{10 \times 40}{60} = 6.67$	$\frac{10 \times 60}{60} = 10$
Forklifting cost per unit →	$0.06 \times 108 = 6.48$	$0.06 \times 106 = 6.36$	$0.06 \times 117 = 7.02$
Supervising cost per unit →	$\frac{10 \times 30}{60} = 5$	$\frac{10 \times 40}{60} = 6.67$	$\frac{10 \times 60}{60} = 10$
Utilities cost per unit →	$1.705 = 8.50$	$1.70 \times 5 = 8.50$	$1.70 \times 6 = 10.20$
Total cost per unit	192.48	243.70	285.72
Number of units	4,000	3,000	2,000
Total costs	7,69,920	7,31,100	5,71,440

(iii) Comments: The difference in the total costs under the two systems is due to the differences in the overheads borne by each of the products. The Activity Based Costs appear to be more precise.

Question - 7

PQR Ltd. has decided to analyze the profitability of its five new customers. It buys soft drink bottles in cases at ₹45 per case and sells them to retail customers at a list price of ₹54 per case. The data pertaining to five customers are given below:

Particulars	Customers				
	A	B	C	D	E
Number of cases sold →	9360	14200	62000	38000	9800
List selling price ₹ →	54	54	54	54	54
Actual selling price → <i>Disc.</i>	54	53.40	49	50.20	48.60
Number of purchase orders →	30	50	60	50	60
Number of customers visits →	4	6	12	4	6
Number of deliveries ✓	20	60	120	80	40
Kilometers travelled per delivery →	40	12	10	20	60
Number of expedite deliveries →	0	0	0	0	2

It's five activities and their cost drivers are:

Activity	Cost Driver
Order taking	→ ₹200 per purchase order
Customer visits	→ ₹300 per each visit
Deliveries	→ ₹4.00 per delivery km travelled
Product Handling	→ ₹2.0 per case sold
Expedited deliveries	→ ₹100 per each such delivery

Net Rev.
 (-) Cost
 Gross Margin

(-) O/S

Op. Income

You are required to:

- Compute the customer level operating income of each of five retail customers by sing the cost driver rates.
- Examine the results to give your comments on Customer 'D' in comparison with Customer 'C' and on Customer 'E' in comparison with Customer 'A'.

Solution**(i) Statement of operating income**

Particulars	Customer A	Customer B	Customer C	Customer D	Customer E
Units →	9,360	14,200	62,000	38,000	9,800
Revenue [54 × No. of units]	5,05,440	7,66,800	33,48,000	20,52,000	5,29,200
(-) Discount → [(List price - Actual price) × No. of units]	-	8,520	3,10,000	1,44,400	52,920
Net revenue ⇒ <i>(Actual SP × units)</i>	5,05,440	7,58,280	30,38,000	19,07,600	4,76,280

(-) Order taking [$200 \times \text{No. of purch. order}$]	6,000	10,000	12,000	10,000	12,000
(-) Customer visit [$300 \times \text{No. of visit}$]	1,200	1,800	3,600	1,200	1,800
(-) Deliveries [$4 \times \text{km travel} \times \text{No. of deliveries}$]	3,200	2,880	4,800	6,400	9,600
(-) Production handling [$2 \times \text{No. of units}$]	18,720	28,400	1,24,000	76,000	19,600
(-) Expedited deliveries [$100 \times \text{No. of delivery}$]	-	-	-	-	200
(-) COGS [$45 \times \text{No. of units}$]	4,21,200	6,39,000	27,90,000	17,10,000	4,41,000
Operating Income	55,120	76,200	1,03,600	1,04,000	(7,920)

(ii) Separate disclosure of revenue helps us to identify the relationship between discount and sales quantity.

<u>Customer</u>	<u>Quantity</u>	<u>Discount</u>	<u>Discount %</u>
A	→ 9,360	-	0% ✓
C	→ 62,000	5	$5 \div 54 = 9.25\%$ ✓
D	→ 38,000	3.80	$3.80 \div 54 = 7.03\%$ ✓
E	8,775	5.40	$5.40 \div 54 = 10\%$

Customer D gets lower discount as compared to Customer C. It may be due to lower quantity purchased by customer D as compared to Customer C.

Customer E gets higher discount as compared to Customer A. Customer E discount is higher in spite of ordering comparative lower quantity and its reason should be further explored.

Question – 8

A drug store is presently selling three types of drugs namely 'Drug A', 'Drug B' and 'Drug C'. due to some constraints, it has decided to go for only one product line of drugs. It has provided the following data for the year 2020-21 for each product line:

		Drug Types		
		A	B	C
Revenue (in ₹)	→	74,50,000	1,11,75,000	1,86,25,000
Cost of goods sold (in ₹)	→	41,44,500	68,16,750	1,20,63,750
Number of purchase orders placed (in nos)		560	810	630
Number of deliveries received		950	1,000	850
Hours of shelf-stocking time		900	1,250	2,350
Units sold (in nos)		1,75,200	1,50,300	1,44,500

Following additional information is also provided:

Activity	Description of Activity	Total Cost (₹)	Cost-allocation base	
Drug License fee	Drug License fee	5,00,000	To be distributed in ratio 2:3:5 between A, B and C	A 60% X
Ordering	Placing of orders for purchases	8,30,000	2,000 purchase orders	415
Delivery	Physical delivery and receipt of goods	18,20,000	2,800 deliveries	650
Shelf stocking	Stocking of goods	32,40,000	4,500 hours of shelf-stocking time	720
Customer Support	Assistance provided to customers	28,20,000	4,70,000 units sold	6
		<u>92,10,000</u>		

You are required to:

- Calculate the operating income and operating income as a percentage (%) of revenue of each product line if:
 - All the support costs (other than cost of goods sold) are allocated in the ratio of cost of goods sold
 - All the support costs (Other than cost of goods sold) are allocated using activity-based costing system.
- Give your opinion about choosing the product line on the basis of operating income as a percentage (%) of revenue of each product line under both the situation as above.

Solution

(i) (a) Statement of operating income

Particulars	Drug A	Drug B	Drug C	Total
Revenue (A)	74,50,000	1,11,75,000	1,86,25,000	3,72,50,000
COGS	41,44,500	68,16,750	1,20,63,750	2,30,25,000
Gross Margin	33,05,500	43,58,250	65,61,250	1,42,25,000
(-) Operating cost (in COGS Ratio)	16,57,800	27,26,700	48,25,500	92,10,000
Operating Income (B)	16,47,700	16,31,550	17,35,750	50,15,000
Operating income % (B ÷ A)	<u>22.12%</u>	<u>14.60%</u>	<u>9.32%</u>	<u>13.46%</u>

(i) (b) Statement of Cost

Particulars	Cost (₹) (A)	Cost Driver (B)	Cost per cost driver (A ÷ B)
Ordering	8,30,000	2,000 purchase order	₹415 per purchase order
Delivery	18,20,000	2,800 deliveries	₹650 per delivery

Shelf stocking	32,40,000	4,500 hours of shelf stocking time	₹720 per hour of shelf stocking time
Customer support	28,20,000	4,70,000 units sold	₹6 per unit sold

Statement of operating income

Particulars	Drug A	Drug B	Drug C
✓ Revenue (A)	74,50,000	1,11,75,000	1,86,25,000
✓ COGS	41,44,500	68,16,750	1,20,63,750
→ Gross Margin (B)	33,05,500	43,58,250	65,61,250
Drug License Fee (in 2:3:5)	1,00,000	1,50,000	2,50,000
Ordering cost	$\frac{415}{2} \times 560 = 2,32,400$	$\frac{415}{3} \times 810 = 3,36,150$	$\frac{415}{5} \times 630 = 2,61,450$
Delivery cost	$\frac{650}{6} \times 950 = 6,17,500$	$650 \times 1000 = 6,50,000$	$650 \times 850 = 5,52,500$
Shelf Stocking cost	$\frac{720}{6} \times 900 = 6,48,000$	$720 \times 1250 = 9,00,000$	$720 \times 2350 = 16,92,000$
Customer support	$\frac{6}{10} \times 175200 = 10,51,200$	$6 \times 150300 = 9,01,800$	$6 \times 144500 = 8,67,000$
Operating cost (C) →	26,49,100	29,37,950	36,22,950
Operating income (B-C=D)	6,56,400	14,20,300	29,38,300
Operating income % (D÷A)	<u>8.81%</u>	<u>12.71%</u>	<u>15.78%</u>

- (ii) When the operating costs are distributed on the basis of cost of goods sold, Drug A has the highest level of operating income percentage because lesser operating cost share is distributed to it.

Activity based costing shows that Drug C uses the large amount of operating cost resources than the other two drugs and simultaneously generates the highest level of revenue and thus operating income percentage is maximum in case of Drug C.